

NETWORKED CONTROL

02/07/2021

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1) Consider the following continuous-time system $\mathcal{S}$

$$\dot{x}(t) = 2x(t) + u(t)$$

- A. Compute the eigenvalue and the mode of the system. Is the system open-loop asymptotically stable?
- B. We aim to control the system using a digital quantized controller. Answer the following questions.
 - a. According to the theory, what is the minimum rate N_B/h (where N_B is the number of bits encoding the control law and h is the sampling time) required to make the system *boundable*?
 - b. In view of the answer given to the previous question, what is the minimum number N_B of bits/packet that is necessary to make the system *boundable* in case $h = \log_e 1.2$ s?
 - c. Set $h = \log_e 1.2$ s and use the value of N_B computed at question 2. Using the control law defined in case we operate at the rate limits (with $u_{min} = -1$ and $u_{max} = 1$), analyze the properties of the state trajectories using the graphical method (possibly starting by different selected initial conditions).
 - d. Is the control law used at point 3. the unique one that allows to steer the state to lie in a bounded set? Why? Compute the corresponding invariant set.

2) What are the different requirements of control networks with respect to data networks?

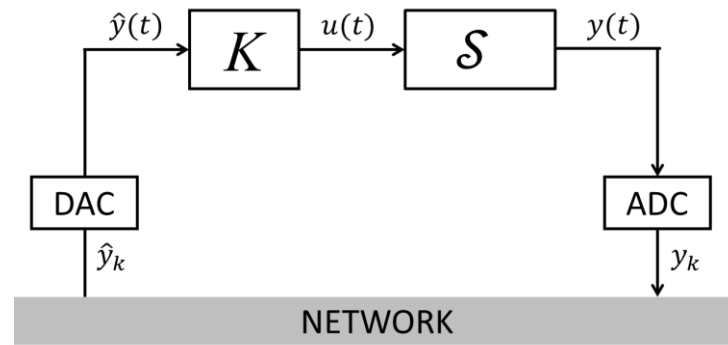
Shortly define data records and data images and comment on the causes of possible differences between the latter.

What issues may arise in case there is a difference between data records and the corresponding data images?

3) Assume that the linear system

$$\mathcal{S}: \begin{cases} \dot{x}(t) &= Ax(t) + Bu(t) \\ y(t) &= Cx(t) \end{cases}$$

is controlled through the networked control system (NCS) sketched in the following figure.



Assume that the NCS is subject to a time-invariant delay τ and to packet dropouts.

- A. Define the asymptotic dropout rate r .
- B. Define the maximal dropout rate and state, in details, the main result allowing to compute it, properly defining all the necessary ingredients.